Software Engineering Lab-1

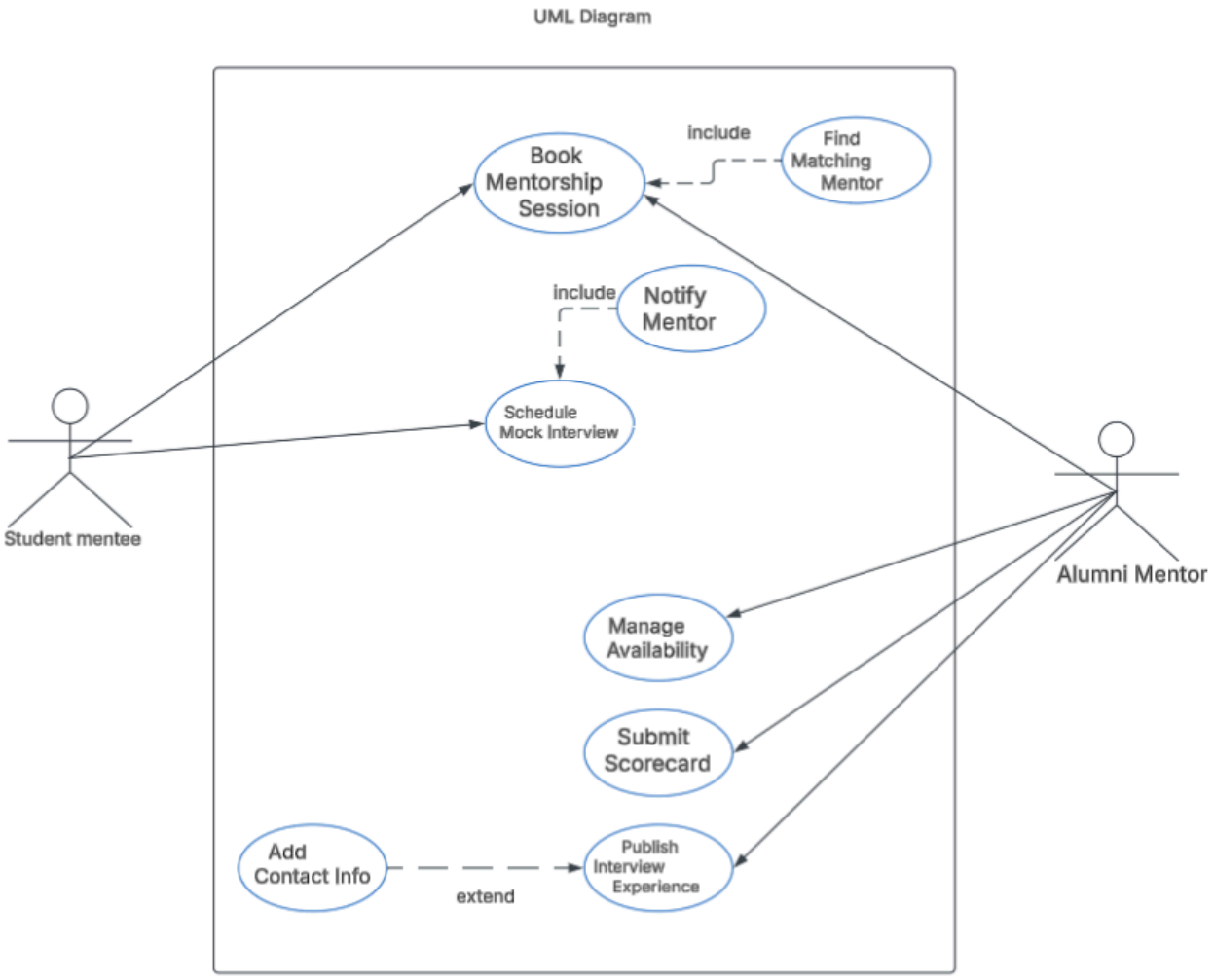
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Assigned Topic Number: 6
Assigned Topic: Alumni Mentorship and Mock Interview Platform

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	**High**	**Pass:** Mentorship session is added to both calendars with a meeting link. **Fail:** Double-booking of mentor slots occurs.	Core matching functionality connecting students to relevant mentors.
FR-002	Functional	The system shall allow alumni mentors to set and update their weekly availability slots on a shared calendar.	**High**	**Pass:** Updated slots appear in student search results within 5 minutes. **Fail:** Stale or already-booked slots are shown as available.	Prevents scheduling conflicts and ensures accurate booking options.
FR-003	Functional	The system shall allow students to schedule mock interview sessions and select an interview domain/track (e.g., DSA, System Design, HR).	**High**	**Pass:** Selected domain is recorded and the mentor is notified with track details before the session. **Fail:** Session is created without a domain tag, or mentor is not notified.	Ensures mentors prepare relevant questions for the chosen track.
FR-004	Functional	The system shall allow mentors to submit a structured post-interview scorecard (ratings + comments) immediately after a session ends.	**Medium**	**Pass:** Scorecard is saved and visible to the student within 10 minutes of session end**.** **Fail:** Scorecard is left blank, or submission is accepted more than 24 hours after session end.	Provides actionable feedback and a performance record for the student.
FR-005	Functional	The system shall allow alumni to publish an "Interview Experience" post containing their company/role, contact info (optional visibility toggle), and the preparation methods/resources that helped them, viewable by students on their profile.	**Medium**	**Pass:** Post is saved, appears on the alumni's public profile, and respects the contact-info visibility toggle chosen by the alumnus. **Fail:** Post is published with contact info visible despite the alumnus opting out, or the post fails to appear after submission.	Gives students access to real, first-hand preparation strategies and optional direct contact.
NFR-001	Nonfunctional	All user profile data and feedback scorecards shall adhere to privacy guidelines and support	**High**	**Pass:** Benchmarking tests confirm target latency and security standards under	Protects sensitive mentee/mentor data

		soft-deletion within 24 hours of request.		simulated peak load. **Fail:** Data remains recoverable/visible after the 24-hour soft-deletion window, or latency exceeds target under load.	and ensures legal compliance.
NFR-002	Nonfunctional	Alumni contact information shared via "Interview Experience" posts shall only be visible to authenticated students and shall never be indexed by external search engines.	**High**	**Pass:** Security audit confirms contact info is inaccessible without login and not crawlable/indexed externally. **Fail:** Contact info is accessible without authentication or appears in external search engine results.	Prevents misuse of alumni personal contact data and supports the optional contact-sharing feature (FR-005).

UML Diagram



Use Case Flow Specification

Use Case: Book Mentorship Session
Actors: Student Mentee (primary), Alumni Mentor

Preconditions

1. The Student Mentee is registered and logged into the platform.
2. At least one Alumni Mentor profile with a matching domain and available time slot exists in the system.
3. The Student Mentee has not already booked a session with the same mentor for an overlapping time slot.

Postconditions

1. A confirmed mentorship session exists, added to both the Student Mentee's and Alumni Mentor's calendars with a meeting link.
2. The booked slot is removed from the mentor's publicly available slots, preventing double-booking.

Main Success Scenario

1. Student Mentee selects "Find a Mentor" and enters a domain of interest (e.g., Cloud Computing).
2. System executes the included use case Find Matching Mentor and returns a ranked list of alumni mentors based on domain match.
3. Student Mentee reviews the recommended mentors and selects one.
4. System displays the selected mentor's available time slots.
5. Student Mentee selects a preferred time slot and clicks "Book Session."
6. System validates that the slot is still available.
7. System reserves the slot, generates a meeting link, and adds the session to both the Student Mentee's and Alumni Mentor's calendars.
8. System sends a confirmation notification to both the Student Mentee and the Alumni Mentor.
9. Use case ends successfully.

Alternate Flow

6a. Slot No Longer Available (Double-Booking Attempt)

6a1. System detects that the selected slot was booked by another student in the Gap.

6a2. System displays a "Slot No Longer Available" message and refreshes the mentor's remaining open slots.

6a3. Student Mentee selects a different available slot and resumes at step 6.

6a4. If no slots remain for that mentor, the Student Mentee is redirected to step 3 to select a different mentor.